

MTH 210 Course Project

Statistical Computing

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Dataset File: fort.58

1 Introduction

The objective of this project is to estimate the parameters of a given probability density function (PDF) using Maximum Likelihood Estimation (MLE) and to construct 95% confidence intervals for these parameters. The provided dataset, `fort.58`, was generated from a distribution where the probability density function for a random variable X is given by:

$$f(x; \alpha, \lambda) = \alpha \lambda e^{-\lambda x} (1 - e^{-\lambda x})^{\alpha-1}; \quad \text{for } x > 0, \text{ and } \alpha, \lambda > 0 \quad (1)$$

To achieve the estimation, a Newton-Raphson (NR) algorithm is implemented, followed by both parametric and non-parametric bootstrap methods for interval construction.

2 Mathematical Results

Given a random sample $x_1, x_2, \dots, x_n$ from the distribution, we know that likelihood function is given as:

$$\mathcal{L}(\alpha, \lambda) = \prod_{i=1}^n f(x_i; \alpha, \lambda) = \alpha \lambda e^{-\lambda x_i} (1 - e^{-\lambda x_i})^{\alpha-1}; \quad \text{for } x > 0, \text{ and } \alpha, \lambda > 0 \quad (2)$$

Taking log of $\mathcal{L}(\alpha, \lambda)$ gives us the log-likelihood function, which is given as

$$\ell(\alpha, \lambda) = \log(\mathcal{L}(\alpha, \lambda)) \quad (3)$$

$$\ell(\alpha, \lambda) = n \ln(\alpha) + n \ln(\lambda) - \lambda \sum_{i=1}^n x_i + (\alpha - 1) \sum_{i=1}^n \ln(1 - e^{-\lambda x_i}) \quad (4)$$

We need to maximize the log likelihood, $\ell(\alpha, \lambda)$ which effectively forms a two dimensional optimization problem. However, we can reduce it to one dimensional optimization by profiling out the parameter α . Consider $\frac{\partial \ell}{\partial \alpha}$ and $\frac{\partial^2 \ell}{\partial \alpha^2}$

$$\frac{\partial \ell}{\partial \alpha} = \frac{n}{\alpha} + \sum_{i=1}^n \ln(1 - e^{-\lambda x_i}) \quad (5)$$

$$\frac{\partial^2 \ell}{\partial \alpha^2} = \frac{-n}{\alpha^2} < 0 \quad \forall \alpha > 0 \quad (6)$$

Setting $\frac{\partial \ell}{\partial \alpha} = 0$ yields $\hat{\alpha}$, which for a given λ , maximizes the log likelihood, that is, $\ell(\alpha, \lambda) \leq \ell(\hat{\alpha}, \lambda)$ for all $\lambda > 0$:

$$\hat{\alpha}(\lambda) = -\frac{n}{\sum_{i=1}^n \ln(1 - e^{-\lambda x_i})} \quad (7)$$

Substituting $\hat{\alpha}(\lambda)$ back into the log-likelihood function reduces the problem to a one-dimensional maximization problem with respect to λ . Let us denote the profile log-likelihood $\ell(\hat{\alpha}(\lambda), \lambda)$ simply as $h(\lambda)$.

$$h(\lambda) = n \ln\left(\frac{-n}{\sum_{i=1}^n \ln(1 - e^{-\lambda x_i})}\right) + n \ln(\lambda) - \lambda \sum_{i=1}^n x_i - \left(n + \sum_{i=1}^n \ln(1 - e^{-\lambda x_i})\right) \quad (8)$$

To apply the Newton-Raphson algorithm, we require the first derivative, $g(\lambda) = h'(\lambda)$, and the second derivative, $g'(\lambda) = h''(\lambda)$.

To simplify the notation for the derivatives, we define the following summation terms based on the sample data:

$$A(\lambda) = \sum_{i=1}^n \frac{x_i e^{-\lambda x_i}}{1 - e^{-\lambda x_i}} \quad (9)$$

$$B(\lambda) = \sum_{i=1}^n \ln(1 - e^{-\lambda x_i}) \quad (10)$$

$$C(\lambda) = \sum_{i=1}^n \frac{x_i^2 e^{-\lambda x_i}}{(1 - e^{-\lambda x_i})^2} \quad (11)$$

Using these defined terms, the first derivative of the profile log-likelihood with respect to λ is:

$$g(\lambda) = \frac{n}{\lambda} - \sum_{i=1}^n x_i - A(\lambda) \left[\frac{n}{B(\lambda)} + 1 \right] \quad (12)$$

Taking the derivative of $g(\lambda)$ with respect to λ yields the second derivative:

$$g'(\lambda) = -\frac{n}{\lambda^2} + C(\lambda) \left[\frac{n}{B(\lambda)} + 1 \right] + A(\lambda)^2 \left[\frac{n}{B(\lambda)^2} \right] \quad (13)$$

These analytical expressions for $g(\lambda)$ and $g'(\lambda)$ are implemented directly in the Newton-Raphson numerical optimization routine to find the MLE for λ , which was then used to find the MLE of α using equation 7.

3 Computational Methodology

3.1 Newton-Raphson Optimization

To find the Maximum Likelihood Estimate (MLE) for λ , we essentially need to find the roots of $g(\lambda) = 0$. The equation can't be solved analytically, and hence we use Newton-Raphson optimization. The algorithm finds the roots using iterative updates. For any iteration step k , if our current estimate is λ_k , the next updated estimate λ_{k+1} is computed using the formula:

$$\lambda_{k+1} = \lambda_k - \frac{g(\lambda_k)}{g'(\lambda_k)} \quad (14)$$

By the definition of the provided probability density function, the parameter λ must reside strictly in the positive domain ($\lambda > 0$). However, an unconstrained Newton-Raphson step can occasionally overshoot and propose a negative value for λ_{k+1} . If such a scenario arises, we apply the step-halving technique, that is, we set $\lambda_{k+1} = \lambda_k/2$. This pulls the estimate back to valid region.

The initial estimate, i.e., λ_0 is in our hands. A poor starting point can cause the algorithm to diverge or get trapped in local optima. For $\alpha=1$, the given distribution essentially becomes an

exponential distribution with parameter λ , for which MLE is inverse of mean. So I took the same for a general α of this distribution. Hence ,

$$\lambda_0 = \frac{1}{\bar{x}} \quad (15)$$

The algorithm then enters an iterative loop (capped at a maximum of 1000 iterations to prevent infinite looping), continually evaluating $g(\lambda)$ and $g'(\lambda)$ at the current estimate to compute the next step.

To ensure the algorithm halts exactly when an optimal solution is reached, we established two distinct stopping criteria:

1. **Derivative Tolerance:** If the absolute value of the gradient becomes extremely close to zero, specifically $|g(\lambda_k)| < 10^{-12}$, we assume we have successfully reached the peak of the log-likelihood curve and terminate the loop.
2. **Step-Size Tolerance:** We also monitor the absolute difference between successive iterations. If the algorithm updates λ by a purely negligible amount, such that $|\lambda_{k+1} - \lambda_k| < 10^{-6}$, we consider the algorithm to have converged and stop.

3.2 Bootstrap Methods for Confidence Intervals

While the Newton-Raphson algorithm provides the point estimates for $\hat{\lambda}$ and $\hat{\alpha}$, it does not directly provide a measure of the variability or precision of these estimates. To quantify this uncertainty and construct 95% confidence intervals, we employ bootstrap resampling techniques. A random seed was set to ensure reproducibility of the generated intervals. We compute the intervals using two distinct approaches: non-parametric and parametric bootstrap, using $B = 1000$ replications for both.

3.2.1 Non-Parametric Bootstrap

The non-parametric bootstrap makes no assumptions about the underlying distribution of the data beyond the empirical distribution observed in the sample.

For each replication $b = 1, 2, \dots, B$:

1. A bootstrap sample, $\mathbf{x}^b = (x_1^b, x_2^b, \dots, x_n^b)$, is generated by randomly sampling n observations from the original dataset $\mathbf{x}$ *with replacement*.
2. The Newton-Raphson optimization routine (as described in Section 3.1) is applied to this newly generated resample $\mathbf{x}^b$.
3. This process yields a new set of maximum likelihood estimates for that specific resample, denoted as $\hat{\lambda}^b$ and $\hat{\alpha}^b$.

After completing $B = 1000$ iterations, we obtain an empirical distribution of the estimators. The 95% confidence intervals are then constructed using the 2.5th and 97.5th quantile of the empirical distribution.

3.2.2 Parametric Bootstrap

Unlike the non-parametric approach, the parametric bootstrap assumes that the original data truly follows the specified PDF, parameterized by the point estimates $\hat{\alpha}_{MLE}$ and $\hat{\lambda}_{MLE}$ computed from the original dataset.

To generate new samples efficiently from this specific distribution, we utilize the Probability Integral Transform. We make the transformation $Y = 1 - e^{-\lambda X}$, the density of Y is then given by:

$$f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right| = \alpha \lambda (1-y) y^{\alpha-1} \cdot \frac{1}{\lambda(1-y)} = \alpha y^{\alpha-1} \quad \text{for } y \in (0, 1) \quad (16)$$

This reveals that $Y \sim \text{Beta}(\alpha, 1)$. Therefore, for each replication $b = 1, 2, \dots, B$:

1. We draw n independent samples $Y_1, \dots, Y_n$ from a $\text{Beta}(\hat{\alpha}_{MLE}, 1)$ distribution.
2. We invert the transformation to compute the parametric simulated data: $x_i^b = -\frac{\ln(1-Y_i)}{\hat{\lambda}_{MLE}}$.
3. We apply the Newton-Raphson optimization routine to this simulated dataset $\mathbf{x}^b$ to obtain $\hat{\lambda}^b$ and $\hat{\alpha}^b$.

Finally, similar to the non-parametric method, the 95% confidence intervals are derived from the 2.5th and 97.5th percentiles of the resulting distribution of estimates.

4 Results

The Newton-Raphson algorithm successfully converged to the global maximum in **5** iterations.

4.1 Maximum Likelihood Estimates

The final point estimates for the parameters are:

- $\hat{\lambda}_{MLE} = \mathbf{2.603779}$
- $\hat{\alpha}_{MLE} = \mathbf{2.67826}$

4.2 95% Confidence Intervals

Based on $B = 1000$ replications, the 95% confidence intervals obtained via the percentile method are:

Method	95% CI for λ	95% CI for α
Non-Parametric Bootstrap	(2.121391, 3.674640)	(1.733191, 6.398400)
Parametric Bootstrap	(1.836909, 3.969332)	(1.703052, 5.958413)